

Selected Formalisms from the Event Density Framework

V. Entanglement, Measurement, and the Commitment Ceiling

Allen Proxmire

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Abstract

Event Density (ED) posits a discrete substrate of *events* fixed by thirteen postulated primitives, with the irreversible *commitment* of events, the arrow of time made physical, as its keystone. Parts I–IV worked gravity, the quantum fields, the substrate’s own theorems, and black holes. This paper works the quantum-foundations sector on a single idea: **commitment individuates**. Before commitment, outcomes are not yet individuated, and that is what entanglement is; the commitment act itself is measurement; and a large enough object commits on its own, capping macroscopic superposition. The paper reproduces the bipartite entanglement formalism, tensor product, Schmidt decomposition, monogamy, no-signaling, von Neumann entropy, and the Bell–Tsirelson structure, and is candid that these are *reproduced, not re-derived*: they are form-forced on a stack of V5 postulates, the Tsirelson bound is inherited, ED adds no new bound, and the irreducibility that defines entanglement is an assertion rather than a theorem. Two results in this half are genuinely strong: no-signaling is over-determined by three independent locks, and, combining the measured exact channel-zero of Part III with the forward arrow, ED delivers locality without superdeterminism. The paper’s distinctive content is the measurement reframe: the pointer basis is einselected, a reconstruction carrying a Solèr residual, no observer required, and because the pointer-basis selector *is* the arrow that is gravity’s time, the collapse variable is *time, not energy*, a deep, structural, but unconfirmed synthesis, with the rate $\tau \propto \hbar/E_G$ reproduced (grounded-in-kind) but E_G identified, not computed. The sector’s one live forward bet is a sharp, architecture-independent ceiling on macroscopic superposition, unifying the matter-wave quantum-classical boundary and the qubit multiplicity walls; its distinctive content is the wall’s *shape*, while the mass scale is a two-point extrapolation now pressured from below by matter-wave interference at ≥ 170 kDa. We state plainly what the sector does not deliver.

1. The substrate, and commitment as individuation

Event Density (ED) is a substrate ontology. Beneath the continuous fields and smooth spacetime of established physics it posits a discrete, graph-like layer of *events*, and treats the familiar continuum laws as coarse-grained readings of that layer. As in Parts I–IV, this paper takes the substrate as given and works one sector, stating at each step what is derived and what is inherited.

Postulated, not derived. The substrate is specified by thirteen primitives, P01–P13 (Paper_087), postulated, not derived, and not claimed minimal. Every result below is a *conditional structural derivation*: given the primitives and the paper-specific postulates named at each step, the stated consequence follows. The keystone is the arrow: P11 (commitment irreversibility) is the substrate event at which a chain’s multi-channel participation resolves to a single channel and no operation returns the post-commitment state to the pre-commitment one, with the one-way character supplied jointly by P11, time homogeneity (P13), and the strictly retarded participation kernel V1 (Theorem T18, Paper_093).

Commitment individuates: the spine of this paper. The keystone does one thing here, and it does it three times. P11 is the act that *individuates* one outcome from many. Read that act at three stages and the whole sector appears:

- *Before* commitment, outcomes are not yet individuated. States are pre-individuated and, in a precise sense, **pre-metric**, the distance between the entangled possibilities is not yet built. This is not “nowhere”: the chains keep their loci; what is unresolved is the individuation, not the location. That regime is **entanglement** (§2–§4).
- The commitment act *itself*, the resolution of a superposition to one channel, is **measurement** (§5).
- And if commitment is a real physical act, a large enough object crosses a threshold and commits *on its own*, which caps how large a superposition can be, the **commitment ceiling** (§6).

One primitive, three hats. And it is the same primitive throughout the series: the arrow that is Part III’s keystone (locality, charge, parity) and Part IV’s horizon-maker is, here, the individuator. This paper is that observation worked out in the quantum-foundations sector.

The honest position, stated up front. This sector reproduces the quantum-mechanical formalism; it does not re-derive it from the primitives, and its value is not a new empirical prediction inside the Bell–Tsirelson correlations. Stated plainly, before any result: the entanglement reproductions are form-forced on six V5 postulates, and the irreducibility that *defines* entanglement is an assertion, not a theorem; the Tsirelson bound is inherited and ED adds no new bound; the measurement reframe, deep as it is, is a structural synthesis that is *unconfirmed* and carries no delivered prediction; and the one live forward bet, the commitment ceiling, is distinctive in its *shape* while its mass number is pressured from below by recent experiment. The paper’s contribution is a *reading* of quantum mechanics in which one primitive does the work of several mysteries, plus one falsifiable ceiling. We say so now, and again in §7. What follows is that reading: entanglement as the pre-commitment regime (§2), the bipartite formalism reproduced (§3), no-signaling and locality (§4), measurement as commitment (§5), and the ceiling (§6).

2. Entanglement as the pre-commitment regime

The first move is interpretive, and it is labeled as such, because it reorganizes the whole sector.

The reframe.

Entanglement, in ED, is the **unresolved-individuation regime**: the substrate has

not yet committed which outcome is which, so the parties are not yet individuated from one another. It is *not* action-at-a-distance. Measurement is the act that joins the entangled possibilities to the committed world; before it, the correlation is real but the individuation is pending (Paper_072).

Tier: A→position (interpretive). Paper_072 is explicit that this reframe *does not modify any quantum-mechanical prediction*; it is a re-description of what the standard formalism already computes, not a new claim about outcomes. We flag it at that tier and lean on it for organization, not for empirical weight.

The reframe earns its keep by dissolving the appearance of spookiness. If entanglement were a channel, the correlations between distant measurements would demand a signal, and the absence of one would be a puzzle. On the pre-individuation reading there is nothing to signal: the correlation is a shared-origin fact about a not-yet-individuated pair, and the “distance” across which it seems to act is a metric that has not been built between them yet. The parties keep their loci (the chains are somewhere), but the *relation* is pre-metric. This is the same common-cause-not-channel structure Part III measured as exact channel-zero, seen from the entanglement side; §4 makes the connection precise.

3. The bipartite formalism, reproduced

The standard machinery of bipartite entanglement all has a substrate reading. Each is a genuine reproduction, and each is form-forced on a named postulate; we give the tier with the result, and we flag the one place the whole arc rests on an assertion.

Tensor product. $\Psi^A B = \Psi^A \otimes \Psi^B$ for a factorizable pair (Paper_063). **Tier: Postulated (P-Bipartite-Mapping).** The tensor-product structure is *part of the definitional commitment* of mapping a bipartite system onto a chain partition, not a separate derivation. Say so: this is the setup, postulated, not a result.

Schmidt decomposition. A bilocal cross-chain state has Schmidt rank greater than one, hence is non-factorizable (Paper_064). **Tier: Derived (standard SVD), conditional on P-V5-Schmidt-Generic; the Schmidt form is inherited** (the singular-value decomposition is standard mathematics). The substrate content is the claim that the V5-bilocal state is *generically* rank > 1 ; that genericity is a postulate, and its derivability from the V5 envelope alone is open.

Monogamy. The Coffman–Kundu–Wootters form of the monogamy inequality (Paper_065). **Tier: form-Derived, conditional on P-V5-Budget + P-Measure-Saturation; the coefficient is inherited.** The form follows from a finite V5 cross-chain budget partitioned additively (P04); that entanglement cannot be freely shared is, in ED, that the budget is finite.

Von Neumann entropy. $S = -\text{Tr} p \log p$ (Paper_068). **Tier: D-via-I (the Shannon–Khinchin uniqueness theorem), conditional on P-Continuity + P-Additivity + P-Maximality; the normalization is inherited.** Two substrate-level derivations of the conditioning postulates are open, and one is *structurally blocked*: P-Maximality would follow from a substrate temperature, and ED has none at this layer. We name that as open, not closed.

Bell–Tsirelson, reproduced exactly. ED reproduces the Bell–Tsirelson–no-signaling nesting and excludes PR-boxes (Paper_069). **Tier: the Tsirelson bound $2\sqrt{2}$ is INHERITED** (the paper states it is “not derived from first principles”), and **the nesting rests entirely on P-V5-Hilbert-Constraint**, which posits the very Hilbert inner product that the Gleason keystone reconstructs only at account tier, so it **inherits a Solèr residual**. The load-bearing sentence: **ED adds no new bound**. It reproduces the quantum correlation set; it does not sharpen or extend it. This is stated here, and again in §4 and §7, because it is the single most important thing not to overclaim in this sector.

The load-bearing honesty: irreducibility is asserted. The property that *defines* entanglement, that the non-factorizability is irreducible (that an entangled state is not secretly a mixture of product states), is, in Paper_063 §3.3, an **A→assertion**: the paper “does not supply an explicit irreducibility proof.” Paper_064 upgrades it to a theorem *only under* P-V5-Schmidt-Generic, a postulate. So the claim the whole arc rests on, entanglement = irreducible non-factorizability, is **postulate-to-postulate, never derived from the primitives**. This is the arc’s soft spot, and it belongs in the body, not buried in an appendix: the bipartite formalism is reproduced form-forced, on a stack of V5 postulates and one unproven irreducibility assertion. It is a coherent structural account, not a primitives-closed derivation.

4. No-signaling, over-determined; and locality without superdeterminism

One result in the entanglement half is unusually clean, and it pairs with a result imported from Part III to give ED a genuine foundational advantage.

No-signaling, over-determined.

The marginal statistics at one wing of a bipartite measurement do not depend on the remote measurement choice, $p(a|x, y) = p(a|x)$, and in ED this is **over-determined by three independent locks** (Paper_066): **L1**, tensor-product structure plus POVM completeness forces marginal independence at the formal level ($\sum_b F_b^y = I_B$); **L2**, the V5 kernel respects substrate causality, so no operational signal crosses a space-like separation even if L1 were somehow evaded; **L3**, the coarse-graining preserves the local-algebra structure. Refuting any one lock leaves the other two intact.

Tier: L1 is D-via-I; L2 rests on P-Substrate-Causality; L3 on the coarse-graining (Paper_073); the “over-determination” verdict is A→position. The paper claims no empirical content beyond standard quantum no-signaling, and neither do we; the interest is structural, that ED forbids signaling three different ways, so no-signaling is not a fine-tuned coincidence but a redundant consequence of the substrate’s architecture. It is the cleanest paper in the arc, and we feature it as such.

Locality without superdeterminism, credited to Part III. No-signaling says no *operational* signal passes. The stronger foundational question is how ED reproduces the Bell/EPR correlations *at all* without either a signal or a hidden-variable conspiracy. The answer combines a Part III result with the arrow. Part III measured that the controlled information capacity between distinct substrate regions is *exactly zero* (the A1 result): correlations are common-cause, never channel. Add to that measured fact two properties of commitment, that it is a *local* substrate event and that its outcome is *stochastic at the moment of commitment*, and the conclusion

follows: ED reproduces the entangled correlations with **no signal channel anywhere and no hidden-variable conspiracy**. This is a genuine advantage over the two standard escapes. A superdeterministic account buys locality by correlating the measurement settings with the hidden state; a signaling account lets one wing influence the other; ED needs neither, because the correlations are common-cause (Part III’s exact zero) and the arrow is forward. We credit the measured locality to Part III and do not re-derive it here; what this sector adds is the reading that the entangled correlation and the strict locality of every committed fact are the two faces of one pre-individuation structure.

5. Measurement as commitment

This is the sector’s conceptual heart, and its most distinctive claim, and we state the claim and its honest tier in the same breath throughout, because the claim is deep and the tier is “unconfirmed.”

The collapse is the commitment. ED has exactly one collapse primitive, and it is P11. A superposition is never itself a channel, so committing a superposed state resolves it to *one* channel, never to the superposition: the channel basis is the unique basis in which outcomes are definite, and it is picked out by the arrow. This is **einselection made primitive rather than derived**. The same irreversibility supplies **repeatability** (a committed channel is locked, so re-measurement returns it with certainty), and because the bandwidth b_K is an intrinsic state property, the **Born probabilities** $P(K) = b_K / \Sigma b$ **are non-contextual** (consistent with Gleason and Kochen–Specker, because outcomes are stochastically committed, not pre-valued). **No observer is required:** einselection is environment-set and coarse ($\gg \ell_{ED}$), so a rock commits; consciousness plays no role. **Tier: a reconstruction of quantum kinematics via the Gleason / Piron–Solèr chain, not a closed theorem;** orthogonality is reduced to operational distinguishability and the complex field $\mathbb{C}$ is selected, with the Solèr lattice rigor the standing residual.

The signature identity: the collapse variable is time, not energy.

ED’s gravity is khronometric, built on a preferred foliation, and that foliation is the arrow made dynamical, the khronon. So the preferred time quantum mechanics wants (to make outcomes definite *at* a time) and the preferred foliation ED’s gravity runs on are **not two structures to reconcile; they are one primitive, coarse-grained into two sectors**. The pointer-basis selector *is* the khronon *is* gravity’s time. Therefore the collapse variable in ED is **time** (the committed arrow), one step upstream of the **energy** E_G that the gravitational-collapse program (Diósi, Penrose) works with.

Tier: structural synthesis, UNCONFIRMED, “not forced by a theorem, carries no delivered prediction.” We use those words. The identification of the einselection arrow with the gravitational khronon is argued from the finding that commitment is the sole process primitive; it is deep and distinctive, and it is not proven. Its payoff is interpretive and real: it *dissolves* the problem of time (QM wants a preferred time, general covariance forbids one) rather than compromising on it, because ED genuinely has a preferred time and it does both jobs; and read against Penrose specifically, whose superposition collapses *because there is no fact*

about which time flows across two superposed spacetimes, ED **supplies exactly the object whose absence drives his argument**. That is a genuine contribution to the framing of the measurement problem. It is not a number.

The rate, honestly. ED gives a *scaling argument* for the collapse rate: the two branches of a superposition source two bandwidth fields, hence two commitment-clocks the single arrow cannot hold, so einselection individuates them at $\tau \propto \hbar/E_G$ (the companion derivation). This **reproduces the Diósi–Penrose scaling with a deeper origin**, and carries two refinements over those rivals, a *definite* reduction (not mere decoherence to a mixture) and a *native* regulator for the point-mass divergence (the finite substrate grain). But E_G **is identified with Penrose’s self-energy, not computed** from the participation fields, and the coefficient and exact energy functional are open. **Tier: grounded-in-kind scaling, not a structural derivation.** ED reproduces the rate and improves its framing; it does not yet deliver the number.

The 2020 bound, and honest de-weaponizing. A 2020 underground experiment excluded the simplest parameter-free Diósi–Penrose model by searching for the spontaneous radiation a continuous collapse must emit. That bound **does not exclude ED**, for a computed reason: ED’s Lindblad jump operators are *channel projectors* ($\hat{J}_\alpha = g_\alpha \hat{\Pi}_\alpha$, forced from P11), so a committed charge is an *exact fixed point* of the dissipative dynamics, with no continuous momentum diffusion and no radiation. **Tier: structural result (forced identification + exact algebra).** But we state the limit as plainly as the result: **this immunity is shared with standard quantum mechanics**; textbook QM predicts the same null. ED’s distinction here is structural (it is a genuine collapse theory that nonetheless sits on the non-radiating side), not a new observable. It is immunity, not a win, and it is filed as such. One prerequisite carries a named open residual: the argument needs ED’s pointer basis to be coarse ($\gg \ell_{ED}$), so that a bound charge is single-channel and the continuum diffusion limit is never reached. That coarseness is **resolved grounded-in-kind** on three standing corpus results (the ℓ_{ED} -fine dynamics is coherent Schrödinger evolution, not measurement; einselection is environment-set; distinguishability is operational, A1), with the explicit channel-granularity theorem and its exact distinguishability floor a remaining write-up, not a delivered proof.

6. The commitment ceiling: the one live forward bet

Every result so far is a reproduction or a reframe. This section is the sector’s one falsifiable forward bet, and, unlike Part IV’s analog-only correction, it is testable in real experiments now.

The ceiling.

If commitment is a real physical act, then as an object’s internal multiplicity grows it crosses a substrate threshold M_{crit} above which P11 individuates the configuration on its own: a **ceiling on macroscopic superposition**. The matter-wave quantum-classical boundary and the qubit-array multiplicity walls are then not two phenomena but two instantiations of crossing the same threshold (Paper_060).

Tier: Grounded (on P-Mcrit-Unified + P-Cross-Platform-Universality); the platform values are inherited, and the comparison to a phase transition is A→analogy, not a derivation. The unification is form-forced: a single substrate threshold, mapped differently onto molecule mass and onto qubit count. That is a real structural claim, and it is

postulate-conditional; we tier it so.

The distinctive claim is the shape. What makes the ceiling a genuine prediction rather than a restatement is its *shape*: a **sharp, architecture-independent** wall (Paper_056). That diverges from the two standing alternatives. Decoherence-only has no sharp wall, only a rate that grows with size; continuous-collapse models (CSL) have a wall but a *tuned* one, with free parameters set to place it. ED's wall is a substrate threshold, sharp and not architecture-tunable. **Tier: Prediction (live, shape-level), and a real-experiment one**; matter-wave interferometry is climbing the mass ladder, and a *sharp architecture-independent* onset (or its absence) is the falsifier.

The number is pressured, and we do not bank it. The mass scale, 140–250 kDa (central ~180), is a **two-point extrapolation** (from interference demonstrated near 10 and 25 kDa), not a substrate-level prediction. And it is under pressure from below: matter-wave interference has **already been demonstrated at ≥ 170 kDa (Nature 2026)**, inside the predicted window and still fully quantum-consistent, which squeezes the number from below. A proposed secondary signature (a second-harmonic feature) is **confounded**, because the interferometer optics generate harmonics regardless. So the honest presentation is: the **shape is the live falsifier** (a sharp, architecture-independent onset, wherever it sits), and the **number is pressured** and must not be banked as a firm prediction. We present both in the same breath, every time.

7. What this sector does not claim

The reproductions and the reframe are coherent, and the disclaimers matter more for that, not less. Stated plainly:

- **Reproduce, not derive.** The bipartite entanglement formalism is *reproduced*, form-forced on six V5 postulates; ED does not derive entanglement from the primitives. The **ir-reducibility that defines entanglement is an A→assertion**, never primitives-derived; and two von Neumann-entropy derivations are open (one structurally blocked, for lack of a substrate temperature).
- **The individuation reframe is interpretive.** The §2 reading of entanglement as the pre-commitment, pre-metric regime is an A→position (Paper_072): it re-describes what the standard formalism already computes and changes no quantum-mechanical prediction.
- **Tsirelson is inherited; no new bound.** The $2\sqrt{2}$ bound is inherited, and the $\text{Bell} \subsetneq \text{Tsirelson} \subsetneq \text{NS}$ nesting rests on P-V5-Hilbert-Constraint plus a Solèr residual. **ED introduces no new empirical content to the Bell/Tsirelson correlations.**
- **The measurement reframe is unconfirmed.** That the einselection arrow *is* the gravitational khronon, so the collapse variable is time not energy, is a **structural synthesis, not a theorem, carrying no delivered prediction**. The rate $\tau \propto \hbar/E_G$ is a grounded-in-kind scaling with E_G **identified, not computed** (coefficient and energy functional open). The 2020-bound immunity is **shared with standard QM** (structural, not an observational win).
- **The ceiling is Grounded, not Derived.** Only its **shape** is distinctive; the **mass number (140–250 kDa) is pressured** from below by ≥ 170 kDa matter-wave interference (Nature 2026), and the second-harmonic secondary signature is confounded. The M_{crit} unification is postulate-conditional and the phase-transition comparison is an analogy.

- **The EFE fingerprint is unmeasurable near Earth.** ED predicts the collapse rate depends on the ambient gravitational field (an anisotropy the continuous-noise models do not carry), but the effect is 10^{-6} -small near Earth and needs a deep-void environment ($g_{ext} \lesssim a_0/100$) to see; mention, do not bank.
- **Nothing forced from nothing.** Every result is conditional on the thirteen primitives and the named V5 postulates, which are themselves postulated, not derived.

8. Appendix: tier table

Every load-bearing claim, with its tier transcribed from the entanglement, state-reduction, and q-compute tiered-claims ledgers and verified against the source paper.

Result	Claim	Tier	Inherited / open
072	Entanglement = unresolved-individuation regime (not action-at-a-distance)	A→position (interpretive)	does not modify QM predictions
063	Tensor product $\Psi^A B = \Psi^A \otimes \Psi^B$	Postulated (P-Bipartite-Mapping)	definitional
064	Schmidt rank $> 1 \implies$ non-factorizable	Derived (SVD) , conditional on P-V5-Schmidt-Generic	Schmidt <i>form</i> inherited (standard SVD)
065	Monogamy (CKW form)	form-Derived , conditional on P-V5-Budget + P-Measure-Saturation	coefficient inherited
068	von Neumann entropy $S = -Tr \rho \log \rho$	D-via-I , conditional on P-Continuity + P-Additivity + P-Maximality	normalization inherited; 2 substrate derivations OPEN (one blocked)
063 §3.3	Irreducibility of non-factorizability	A→assertion	never primitives-derived; the arc's soft spot
066	No-signaling $p(a\ x, y) = p(a\ x)$, over-determined (3 locks)	D-via-I (L1) + Grounded (L2) + I (L3)	"over-determination" = A→position; the cleanest paper
069	Tsirelson bound $2\sqrt{2}$	Inherited	"not derived... INHERITED"; ED adds no new bound

Result	Claim	Tier	Inherited / open
069	Bell $\not\subseteq$ Tsirelson $\not\subseteq$ NS nesting; PR-box exclusion	D-via-I , conditional on P-V5-Hilbert- Constraint	Solèr residual (Gleason keystone, account-tier)
positioning	Einselection primitive; pointer basis = arrow-selected; $\mathbb{C}$ selected	Reconstruction (Gleason / Piron–Solèr)	Solèr lattice residual OPEN
positioning	Arrow = gravity’s time = pointer-basis selector $\Rightarrow$ collapse variable = time, not energy	Structural synthesis, UNCONFIRMED	not forced by a theorem; no delivered prediction
CollapseRate	Collapse rate $\tau \propto \hbar/E_G$	Grounded-in-kind scaling	E_G identified, not computed ; coefficient + energy functional OPEN
CollapseRate	2020-bound immunity (jump ops = channel projectors $\Rightarrow$ committed states = exact fixed points)	Structural result (FORCED)	shared with standard QM ; immunity, not a win
positioning	Locality without superdeterminism (A1 + forward arrow)	Grounded (A1 measured)	cross-ref Part III (A1)
060	M_{crit} unification (matter-wave Q-C boundary + qubit walls = one threshold)	Grounded (P-Mcrit-Unified + P-Cross-Platform- Universality)	values inherited; phase-transition = A $\rightarrow$ analogy
056	Class-A ceiling shape (sharp, architecture- independent)	Prediction (live, shape-level)	genuinely distinctive (vs decoherence-only + CSL)
056	Class-A ceiling number (140–250 kDa)	Selected/Inherited; PRESSURED	2-point extrapolation; ≥ 170 kDa matter-wave (Nature 2026) inside window, still QM-consistent; 2nd-harmonic CONFOUNDED; do not bank

Result	Claim	Tier	Inherited / open
OpenPiece-i	Collapse rate depends on ambient field (EFE), anisotropic	Prediction (distinctive-in- principle)	10^{-6} near Earth, unmeasurable (needs deep-void $g_{ext} \lesssim a_0/100$)

Series context. Part V of the *Selected Formalisms* series, after gravity (I), quantum fields (II), the substrate's own theorems (III), and black holes (IV). This is the quantum-foundations sector, worked on one idea, commitment individuates: entanglement is the pre-commitment regime, measurement is the commitment act, and a large enough object commits on its own. The bipartite formalism and the Bell–Tsirelson structure are reproduced form-forced (Tsirelson inherited, no new bound, the defining irreducibility asserted); the distinctive content is the reframe (the collapse variable is time, not energy, because the pointer-basis selector is the arrow that is gravity's time) and one live forward bet (the commitment ceiling's shape). One primitive, P11, the same arrow that is Part III's keystone and Part IV's horizon-maker, does all three jobs; the sector's price, and its coherence, are stated as plainly as its results.